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United States Patent Application Publication

20250269422

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

NISHINO; YOSHIRO et al.

JOINING METHOD, MANUFACTURING METHOD, AND IMAGE FORMING APPARATUS

Abstract

A joining method of joining a first metal sheet and a second metal sheet, the joining method including a fastening step of fastening the first metal sheet and the second metal sheet, which have been superposed on each other, by a fastening member at a fastening position, and a welding step of welding the first metal sheet and the second metal sheet, which have been superposed on each other, at a welding position, wherein, in a direction intersecting a sheet thickness direction of the first metal sheet, a distance between the fastening position and the welding position is set such that vapor generated between the first metal sheet and the second metal sheet during the welding step elastically deforms at least one of the first metal sheet and the second metal sheet.

Inventors: NISHINO; YOSHIRO (Chiba, JP), CHIBA; HIROSHI (Ibaraki, JP), KITAJIMA; TOMOHARU (Ibaraki, JP), MOTEGI; JUNICHI (Chiba, JP), IWANAGA; AYUMU (Chiba, JP), KAKIZAKI; SHIMPEI (Chiba, JP)

Applicant: CANON KABUSHIKI KAISHA (Tokyo, JP)

Family ID: 1000008477478

Appl. No.: 19/057812

Filed: February 19, 2025

Foreign Application Priority Data

JP	2024-025599	Feb. 22, 2024
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Publication Classification

Int. Cl.: B21J15/02 (20060101); B41J29/02 (20060101); G03G21/16 (20060101)

U.S. Cl.:

Background/Summary

BACKGROUND

Field

[0001] The present disclosure relates to a welding technology for a metal sheet with metallicity, such as a frame member to be used in manufacture of a frame body of an apparatus such as an image forming apparatus.

Description of the Related Art

[0002] A frame body of an electrophotographic, ink-jet, or other image forming apparatus is formed by combining a plurality of frame members. As a joining method of fixing frame members together, joining with screw fixation, rivet joining, joining by welding, and the like are known. The joining method of fixing frame members together by welding can provide higher rigidity of the frame body structure as compared to other joining methods, and can secure stable positioning accuracy of the frame members, and hence the joining method by welding has been widely used. When joining by welding is performed, the frame members are heated during welding. This heat may cause distortion of the frame members.

[0003] In order to minimize distortion of the frame members, it is preferred that the amount of heat to be applied to the frame members be suppressed as much as possible. As a joining method of suppressing the amount of heat to be applied to the frame members, there is known a superposition joining method in which two frame members to be joined are arranged so that surfaces of the two frame members are partially superposed on each other, and the superposed surfaces are irradiated with a laser beam, thereby joining the two frame members together. This joining method has been often used as a joining method of joining frame members of a frame body of an image forming apparatus.

[0004] Galvanized steel sheets are often used for frame members forming a frame body of an image forming apparatus in order to secure rust preventiveness. When superposition welding is performed using steel sheets plated with a metal that has a boiling point lower than a melting point of a base metal sheet, such as galvanized steel sheets, vapor of the vaporized plating metal may remain between the superposed surfaces in close contact. The vapor of the plating metal creates blowholes in the fused and joined portion, and thus causes deterioration in welding quality.

[0005] Accordingly, in a jig or the like that superposes and supports frame members during welding, a pressing portion that presses the superposed surfaces so as to prevent the superposed surfaces from separating from each other is formed of an elastic member. In such a configuration, when metal vapor is generated, the pressing portion moves slightly, and thus a minute gap is defined between the members. The metal vapor is released from this minute gap between the members. The frame body of an image forming apparatus has a configuration in which a large number of frame members are mounted from a plurality of directions. Thus, the jig that supports the frame members is configured so as to cover the frame body, and hence is often large in scale.

[0006] Image forming apparatus are becoming larger in size along with an increase in speed. Further, there are an increasing number of image forming apparatus in which a plurality of modules having functions related to image formation are mounted on a plurality of frame bodies in a divided manner. The plurality of frame bodies have different modules to be mounted thereon, and hence differ in frame body structure. When large-scale jigs are used for welding the plurality of frame bodies, a plurality of large-scale jigs are preferred for the plurality of frame bodies, respectively, resulting in, for example, huge investment in jigs and huge man-hours required for setup and changeover during production.

[0007] As a joining method for a frame body, which does not use a large-scale jig, there is known a method of restraining the superposed surfaces from separating from each other by a fastening member such as a rivet as a temporary support for the frame members before welding, and then welding the frame members under a state in which the frame body is allowed to stand on its own without using a jig. However, in a case where the superposed surfaces of the frame members are temporarily supported by the fastening member, the fastening member has no elasticity, and hence a minute gap is not formed between the superposed surfaces of the frame members at the time when metal vapor is generated. Thus, it is difficult to release the metal vapor. Japanese Patent Application Laid-open No. 2014-94390 discloses a joining method in which a gap is defined by providing a step at a joining point formed by welding, to thereby form a ventilation channel for releasing the vapor of the plating metal. In this joining method, the gap is defined by a shape of the frame member. Accordingly, even after the frame members are temporarily supported, the ventilation channel is secured, and hence the metal vapor is removed.

[0008] However, as described above, the frame body of an image forming apparatus has the configuration in which a large number of frame members are mounted from a plurality of directions, and hence there are a large number of joining points at which the frame members are welded and joined together, and the joining points face in a plurality of directions. Accordingly, in a case where a gap is defined between the superposed surfaces by the shape of the frame member to secure the ventilation channel for the vapor of the plating metal, it is preferable to provide a step shape for each of a large number of welding points, resulting in complex shapes of the frame members. That is, in order to manufacture a frame body by intricately combining a plurality of frame members, the number of manufacturing steps for a frame member increases, and the number of faces of a die and manufacturing man-hours increase.

SUMMARY

[0009] A joining method of joining a first metal sheet and a second metal sheet according to some embodiments of the present disclosure includes a fastening step of fastening the first metal sheet and the second metal sheet, which have been superposed on each other, by a fastening member at a fastening position, and a welding step of welding the first metal sheet and the second metal sheet, which have been superposed on each other, at a welding position, wherein, in a direction intersecting a sheet thickness direction of the first metal sheet, a distance between the fastening position and the welding position is set such that vapor generated between the first metal sheet and the second metal sheet during the welding step elastically deforms at least one of the first metal sheet and the second metal sheet to form a gap between the first metal sheet and the second metal sheet. A joining method of joining a first metal sheet and a second metal sheet according to another embodiment of the present disclosure includes a fastening step of fastening the first metal sheet and the second metal sheet, which have been superposed on each other, by a fastening member at a fastening position, and a welding step of welding the first metal sheet and the second metal sheet, which have been superposed on each other, at a welding position, wherein a distance between the fastening position and the welding position in a direction intersecting a sheet thickness direction of the first metal sheet is 30 mm or more.

[0010] A manufacturing method of an image forming apparatus according to yet another embodiment of the present disclosure includes preparing a frame body using a joining method of joining a first metal sheet and a second metal sheet including a fastening step of fastening the first metal sheet and the second metal sheet, which have been superposed on each other, by a fastening member at a fastening position, and a welding step of welding the first metal sheet and the second metal sheet, which have been superposed on each other, at a welding position, wherein, in a direction intersecting a sheet thickness direction of the first metal sheet, a distance between the fastening position and the welding position is set such that vapor generated between the first metal sheet and the second metal sheet during the welding step elastically deforms at least one of the first metal sheet and the second metal sheet to form a gap between the first metal sheet and the second

metal sheet. A manufacturing method of an image forming apparatus according to yet another embodiment of the present disclosure includes preparing a frame body using a joining method of joining a first metal sheet and a second metal sheet including a fastening step of fastening the first metal sheet and the second metal sheet, which have been superposed on each other, by a fastening member at a fastening position, and a welding step of welding the first metal sheet and the second metal sheet, which have been superposed on each other, at a welding position, wherein a distance between the fastening position and the welding position in a direction intersecting a sheet thickness direction of the first metal sheet is 30 mm or more.

[0011] An image forming apparatus according to yet another embodiment of the present disclosure includes an image forming unit, and a frame body, which includes a first metal sheet and a second metal sheet joined together, and supports the image forming unit, wherein the first metal sheet and the second metal sheet are fastened together by a fastening member at a fastening position, wherein the first metal sheet and the second metal sheet are welded together at a welding position, and wherein a distance between the fastening position and the welding position in a direction intersecting a sheet thickness direction of the first metal sheet is 30 mm or more.

[0012] Further features of the present disclosure will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 is an explanatory view for illustrating a joining method.

[0014] FIG. 2 is an explanatory view for illustrating the joining method.

[0015] FIG. 3 is a flowchart for illustrating the joining method.

[0016] FIGS. 4A, 4B, 4C, 4D, and 4E are each a schematic sectional view for illustrating a region near a joining point during welding.

[0017] FIG. 5 is an explanatory graph for showing a relationship between a gap and tensile strength of a steel sheet.

[0018] FIG. 6 is an explanatory graph for showing a relationship between a distance and the tensile strength.

[0019] FIG. 7 is an explanatory graph for showing a relationship between the distance and a rate of occurrence of blowholes.

[0020] FIG. 8 is a configuration view for illustrating an ink-jet recording apparatus.

[0021] FIG. 9 is a configuration view for illustrating an example of a frame body of a fixing module.

[0022] FIG. 10 is a configuration view for illustrating an example of a bottom plate frame.

[0023] FIG. 11 is an exploded view for illustrating a fixing frame.

[0024] FIG. 12 is a configuration view for illustrating a state in which each frame of the fixing frame is temporarily supported.

[0025] FIG. 13 is an enlarged view for illustrating a region B of FIG. 12.

[0026] FIG. 14 is an explanatory view for illustrating a state in which the fixing frame before a joining step is mounted on a welding apparatus.

[0027] FIG. 15 is an enlarged view for illustrating the region B of FIG. 12 after joining.

[0028] FIG. 16 is an explanatory view for illustrating a related-art temporary support jig for welding a frame body.

[0029] FIGS. 17A, 17B, and 17C are each an explanatory view for illustrating a related-art frame joining method.

[0030] FIG. 18 is an explanatory view for illustrating a state in which the temporary support jig supports each frame therein.

[0031] FIG. **19** is an explanatory view for illustrating a state in which the temporary support jig is mounted on the welding apparatus.

DESCRIPTION OF THE EMBODIMENTS

[0032] Now, referring to the accompanying drawings, description is given of various exemplary embodiments, features, and aspects of the present disclosure. Regarding a joining method and a frame structure of an image forming apparatus as described in this embodiment, dimensions, materials, shapes, relative arrangements, and the like of component parts as described are not intended to limit the scope of the present disclosure only thereto, unless otherwise specifically stated.

Joining Method

[0033] FIG. **1** and FIG. **2** are each an explanatory view for illustrating a joining method according to this embodiment. Here, description is given of a method of welding galvanized steel sheets, which are metal sheets with metallicity to be joined. FIG. **1** is a top view for illustrating galvanized steel sheets joined by the joining method according to this embodiment. FIG. **2** is a schematic sectional view for illustrating a region near a joining point.

[0034] A first galvanized steel sheet **101** and a second galvanized steel sheet **102** are arranged so as to partially superposed on each other. A hole **105** is formed in the first galvanized steel sheet **101**, a hole **106** is formed in the second galvanized steel sheet **102**, and a rivet **103** is provided through the hole **105** and the hole **106**. The rivet **103** fastens the first galvanized steel sheet **101** and the second galvanized steel sheet **102** so as to prevent superposed surfaces of the first galvanized steel sheet **101** and the second galvanized steel sheet **102** from separating from each other. A joining point **104** (welding position) formed by laser welding is present at a position apart by a predetermined distance “d” from a center of the rivet **103** (fastening position) in a direction intersecting a sheet thickness direction of the first galvanized steel sheet **101** (or second galvanized steel sheet **102**). On the second galvanized steel sheet **102**, a protrusion **111** is formed as a mark of laser welding.

[0035] FIG. **3** is a flowchart for illustrating the joining method of performing such laser welding. FIGS. **4A** to **4E** are each a schematic sectional view for illustrating a region near the joining point during welding.

[0036] When laser welding is started, first, a part of the first galvanized steel sheet **101** and a part of the second galvanized steel sheet **102** are superposed so as to be in close contact with each other (Step **S101**). FIG. **4A** shows, as an example, the first galvanized steel sheet **101** and the second galvanized steel sheet **102** during a superposing step. The hole **105** formed in the first galvanized steel sheet **101** and the hole **106** formed in the second galvanized steel sheet **102** are aligned so as to be substantially coaxial with each other.

[0037] After the superposing step, the first galvanized steel sheet **101** and the second galvanized steel sheet **102** are fastened to each other as temporary support so that the superposed surfaces of the first galvanized steel sheet **101** and the second galvanized steel sheet **102** are prevented from separating from each other in the vertical direction (Step **S102**). Specifically, as illustrated in FIG. **4B**, the rivet **103** is inserted as a fastening member through the hole **105** and the hole **106** that have been aligned so as to be substantially coaxial with each other. The rivet **103** fastens the first galvanized steel sheet **101** and the second galvanized steel sheet **102** to each other. The fastening step prevents contact surfaces of the first galvanized steel sheet **101** and the second galvanized steel sheet **102** from separating from each other.

[0038] After the fastening step, the first galvanized steel sheet **101** and the second galvanized steel sheet **102** are joined to each other by welding (Step **S103**). In the joining step, specifically, as illustrated in FIG. **4C**, a laser beam **107** output from a laser welding apparatus (not shown) is radiated to a position apart by a predetermined distance “d” from the center of the rivet **103** (fastening position). The radiation of the laser beam **107** forms a molten pool **108** in the first galvanized steel sheet **101** and the second galvanized steel sheet **102** as illustrated in FIG. **4D**. When the molten pool **108** solidifies, the first galvanized steel sheet **101** and the second galvanized

steel sheet **102** are joined to each other. A position irradiated with the laser beam **107** is the joining point **104** between the first galvanized steel sheet **101** and the second galvanized steel sheet **102**. [0039] With the irradiation with the laser beam, both the first galvanized steel sheet **101** and the second galvanized steel sheet **102** generate vapor of zinc, which is a plating metal. This is because a boiling point of zinc, which is a metal to be used as a coating material for coating the surfaces of the first galvanized steel sheet **101** and the second galvanized steel sheet **102**, is lower than a melting point of a steel sheet (iron) being a base metal. In a case where the surfaces of the first galvanized steel sheet **101** and the second galvanized steel sheet **102** cannot be separated from each other between the superposed surfaces thereof, zinc vapor remains in the molten pool **108** because there is no place for the zinc vapor to escape. This causes, for example, blowholes to occur. [0040] In this embodiment, the joining point **104** is at a position apart by a predetermined distance “d” from the rivet **103** that restrains the first galvanized steel sheet **101** and the second galvanized steel sheet **102** from separating from each other. Accordingly, when zinc vapor is generated between the superposed surfaces of the first galvanized steel sheet **101** and the second galvanized steel sheet **102**, at least one of the first galvanized steel sheet **101** and the second galvanized steel sheet **102** can deflect due to elastic deformation (FIG. 4D). In FIG. 4D, a steel sheet portion of the base metal of the first galvanized steel sheet **101** is melted by the laser beam **107** to form the molten pool **108**. When zinc vapor is generated, a portion, on a distal end side from the molten pool **108**, of the first galvanized steel sheet **101** is lifted due to the zinc vapor, thereby causing the first galvanized steel sheet **101** to deflect. Thus, a gap **109** is formed between the superposed surfaces during a period in which welding is performed. The zinc vapor can be released from the gap **109** as an exhaust air **110**.

[0041] When the joining step is completed, as illustrated in FIG. 4E, the protrusion **111** is formed on the surface of the first galvanized steel sheet **101** or the second galvanized steel sheet **102** by welding. The protrusion **111** is not necessarily formed on both of the first galvanized steel sheet **101** and the second galvanized steel sheet **102**, and may be formed on any one of the first galvanized steel sheet **101** and the second galvanized steel sheet **102**. Alternatively, the surface of the first galvanized steel sheet **101** or the surface of the second galvanized steel sheet **102** may be dented at the joining point **104**. The gap **109** formed due to elastic deformation becomes smaller as the shape of first galvanized steel sheet **101** or second galvanized steel sheet **102** that has been deformed is restored to the original shape after the joining step. However, the molten pool **108** may flow into the gap **109**, and in this case, the shape of the gap **109** may not be restored to a shape before the joining step.

Tensile Strength at Joining Point

[0042] FIG. 5 is an explanatory graph for showing a relationship between the gap **109** and the tensile strength of the steel sheets when the first galvanized steel sheet **101** and the second galvanized steel sheet **102** are superposed and welded to each other. As shown in FIG. 5, the weld strength is significantly reduced in a case where the gap **109** between the first galvanized steel sheet **101** and the second galvanized steel sheet **102** exceeds 0.4 millimeter (mm). For this reason, preferably, the gap **109** formed between the superposed surfaces of the first galvanized steel sheet **101** and the second galvanized steel sheet **102** does not exceed 0.4 mm.

[0043] FIG. 6 is an explanatory graph for showing a relationship between the distance “d” from the center of the rivet **103** (fastening position) being the fastening member to the joining point **104** formed by welding, and the tensile strength. The rivet **103** restrains the separation between the first galvanized steel sheet **101** and the second galvanized steel sheet **102**. Further, the tensile strength at the joining point **104** tends to decrease significantly as the distance “d” from the center of the rivet **103** to the joining point **104** becomes smaller.

[0044] FIG. 7 is an explanatory graph for showing a relationship between the distance “d” from the center of the rivet **103** (fastening position) to the joining point **104** and a rate of occurrence of blowholes. The rate of occurrence of blowholes tends to increase significantly as the distance “d”

becomes smaller. From FIG. 7, it is conceivable that the phenomenon described with reference to FIG. 6, in which the tensile strength at the joining point **104** decreases as the distance “d” from the center of the rivet **103** to the joining point **104** becomes smaller, is mainly caused by the occurrence of blowholes. Accordingly, by setting the distance “d” between the center of the rivet **103** and the joining point **104** to a predetermined distance or more, the occurrence of blowholes can be suppressed and the tensile strength can be stabilized more.

[0045] In this embodiment, description is given of an example of welding, as an article to be joined, a thin sheet formed of a galvanized steel sheet, which is mainly used for a frame body of an image forming apparatus. The galvanized steel sheet is assumed to have a sheet thickness of from 0.8 mm to 3.2 mm. The target tensile strength at the joining point formed by welding is 1500 newtons (N) or more. Accordingly, the distance “d” from the center of the rivet **103** to the joining point **104** is set to 30 mm or more with respect to the largest sheet thickness of 3.2 mm.

[0046] Considering a relationship between deflection of a beam and a sectional secondary moment, it can be seen that the distance “d” is proportional to the sheet thickness of the galvanized steel sheet. Accordingly, it is preferred that the distance “d” be at least 9.3 times the sheet thickness. In this case, the sheet thickness is the sheet thickness of the steel sheet of one of the first galvanized steel sheet **101** and the second galvanized steel sheet **102**, which has a larger amount of deflection when the gap **109** is formed. Normally, the sheet thickness in this case is the sheet thickness of the first galvanized steel sheet **101**. A position at which the tensile strength at the joining point **104** is not much needed, the distance “d” from the center of the rivet **103** to the joining point **104** may be less than 9.3 times the sheet thickness.

[0047] In a case where the distance from the rivet **103** is excessively large, the gap between the steel sheets tends to increase due to deflection caused by the self-weight of the galvanized steel sheet, warpage during processing, and the like. Accordingly, in this embodiment, the distance “d” is 250 mm or less so that the gap between the steel sheets is 0.4 mm or less. Similarly to the foregoing, this configuration may not apply to the position at which the tensile strength at the joining point is not much needed.

[0048] In the joining method according to this embodiment, it is not necessary to provide a part shape such as a step for the release of zinc vapor when superposition welding in which plate-like steel sheets (metal sheets) are superposed and welded to each other by laser welding is performed. Further, the joining method according to this embodiment does not need, for example, a step of performing laser irradiation twice, such as performing main welding after removing zinc from the welded surface by laser beam irradiation, or a jig capable of forming a ventilation channel for releasing zinc vapor. In the joining method according to this embodiment, two thin steel sheets are temporarily supported by the rivet **103**, and then joined to each other by radiating a laser beam to a position apart by a predetermined distance “d” from the fastening position of the rivet **103**. In this way, in this embodiment, it is possible to perform welding that prevents deterioration of welding quality due to, for example, occurrence of blowholes.

[0049] In this embodiment, description has been given of the case in which zinc is used as a plating material with a boiling point lower than the melting point of the steel sheet being the base metal, but the plating material is not limited thereto. Description has been given of the case in which the rivet is used as the fastening member for suppressing separation in a plane direction before the welding step, but the fastening member is not limited thereto as long as the fastening member suppresses separation in the plane direction. It is only preferable that the steel sheet from which zinc vapor is generated be at least one of the first galvanized steel sheet **101** and the second galvanized steel sheet **102**. Welding is performed by heating with a laser beam. However, the heating method is not limited to the laser beam, and may be any other known heating method in the art of welding (for example, electricity, arc discharge, gas, plasma, and electron beam).

Operational Form

[0050] The frame body structure of an image forming apparatus using the joining method

according to this embodiment is described. Here, an ink-jet recording apparatus is described as an example of the image forming apparatus.

[0051] FIG. **8** is a configuration view for illustrating an ink-jet recording apparatus. An ink-jet recording apparatus **100** according to this embodiment is a sheet-fed image forming apparatus that forms an ink image on a sheet using two liquids, specifically, a reaction liquid and an ink to produce a printed product. The ink-jet recording apparatus **100** includes a sheet feeding module **1000**, a printing module **2000**, a drying module **3000**, a fixing module **4000**, a cooling module **5000**, a reversing module **6000**, and a sheet discharging and stacking module **7000**. A cut paper-like sheet on which an image is to be printed is fed from the sheet feeding module **1000**, subjected to predetermined processing by each module, and discharged to the sheet discharging and stacking module **7000**.

[0052] The sheet feeding module **1000** includes a plurality of sheet storage portions **1100a** to **1100c** (three tiers in this embodiment). Each of the sheet storage portions **1100a** to **1100c** can store sheets therein. Each of the sheet storage portions **1100a** to **1100c** is configured to be capable of being drawn toward a front side of the apparatus, and sheets are stored in each sheet storage portion after the sheet storage portion is drawn toward the front side of the apparatus. The sheet feeding module **1000** feeds sheets one by one to the printing module **2000**. Thus, each of the sheet storage portions **1100a** to **1100c** includes a separation belt and conveyance rollers. The number of the sheet storage portions **1100a** to **1100c** is an example, and may be one, two, or four or more.

[0053] The printing module **2000** is an image forming unit that forms an image on a sheet fed from the sheet feeding module **1000**. The printing module **2000** includes a pre-image-forming registration correction unit (not shown), a print belt unit **2200**, and a recording unit **2300**. The pre-image-forming registration correction unit corrects an inclination and a position of the sheet fed from the sheet feeding module **1000**, and conveys the sheet to the print belt unit **2200**.

[0054] The print belt unit **2200** and the recording unit **2300** are arranged on a downstream side of the pre-image-forming registration correction unit in a sheet conveying direction so as to be opposed to each other across a sheet conveyance path. The print belt unit **2200** conveys, through suction, a sheet to be conveyed from the pre-image-forming registration correction unit. The recording unit **2300** is a sheet processing unit that performs recording processing (printing) from above by a recording head to form an image on the sheet conveyed by the print belt unit **2200**. The recording head performs printing by ejecting the ink onto the sheet. The sheet is conveyed through suction by the print belt unit **2200**, and thus a constant clearance between the recording head and the sheet is maintained.

[0055] A plurality of recording heads are arranged along the sheet conveying direction. The recording heads in this embodiment are five line-type recording heads corresponding to four colors of Y (yellow), M (magenta), C (cyan), and K (black), and the reaction liquid. The number of colors and the number of recording heads are not limited to five. As the ink-jet system, a system using a heat generating element, a piezoelectric element, an electrostatic element, or a micro-electromechanical system (MEMS) element can be adopted. The ink of each color is supplied to the recording head from an ink tank (not shown) through an ink tube.

[0056] The sheet printed in the recording unit **2300** is conveyed by the print belt unit **2200**. An in-line scanner (not shown) is arranged on a downstream side of the recording unit **2300** in the conveying direction. The in-line scanner is used in order to detect misalignment and color density of the image formed on the sheet and to correct an image to be printed.

[0057] The drying module **3000** dries the sheet on which the image has been formed by the printing module **2000**. The drying module **3000** dries the sheet to reduce a liquid component contained in the ink, thereby improving fixability between the sheet and the ink. The drying module **3000** includes a decoupling unit **3200**, a drying belt unit **3300**, and a warm air blowing unit **3400**.

[0058] The sheet printed in the recording unit **2300** of the printing module **2000** is conveyed to the decoupling unit **3200** in the drying module **3000**. The decoupling unit **3200** weakly holds and

conveys the sheet by air pressure from above and belt friction. This prevents displacement of the remaining portion of the sheet on the print belt unit **2200** under a state in which the sheet extends astride the decoupling unit **3200** and the print belt unit **2200**.

[0059] The sheet conveyed from the decoupling unit **3200** is conveyed through suction to the drying belt unit **3300**, and at the same time, hot air is blown to the sheet from the warm air blowing unit **3400** arranged above the belt to dry the ink application surface (image printed surface). In addition to the hot air blowing method, the drying method may include a combination of a method of irradiating the sheet surface with electromagnetic waves (such as ultraviolet rays and infrared rays) and a conduction heat transfer method through contact of heating elements.

[0060] The fixing module **4000** fixes the image on the sheet by heating the sheet dried in the drying module **3000** and by drying the ink. The fixing module **4000** includes a fixing belt unit **4100** including an upper belt unit **4110** and a lower belt unit **4120**. The fixing module **4000** causes the sheet conveyed from the drying module **3000** to pass between the heated upper belt unit **4110** and lower belt unit **4120**, and thus causes (fixes) the ink solvent to fully permeate the sheet.

[0061] The cooling module **5000** cools the sheet on which the image has been fixed by the fixing module **4000**, and thus solidifies the ink softened by heating and suppresses a temperature change of the sheet caused by downstream devices. The cooling module **5000** includes a plurality of cooling units **5001**. The plurality of cooling units **5001** cool the high-temperature sheet conveyed from the fixing module **4000**. Each cooling unit **5001** is configured to cool a sheet by taking outside air into a cooling box with a fan to increase the pressure in the cooling box, and by exposing the sheet to air blown out from nozzles formed in a conveyance guide. The plurality of cooling units **5001** are arranged on both sides of the conveyance path, and hence can cool the sheet from both sides.

[0062] A conveyance path switching unit is provided in the cooling module **5000**. The conveyance path switching unit switches conveyance paths for the sheet according to whether the sheet is conveyed to the reversing module **6000** or to a double-sided printing conveyance path to be used for double-sided printing.

[0063] During double-sided printing, the sheet is conveyed to the lower conveyance path in the cooling module **5000**, and is conveyed through the double-sided printing conveyance paths of the fixing module **4000**, the drying module **3000**, the printing module **2000**, and the sheet feeding module **1000**. A double-sided printing conveyance unit of the fixing module **4000** is provided with a first reversing unit **4200** that reverses the front and back surfaces of the sheet. After the sheet is once conveyed to the first reversing unit **4200**, the sheet is reversed and conveyed to the drying module **3000** side, and thus the printed surface with the image is reversed. Owing to conveyance of the sheet through the first reversing unit **4200**, printing can be performed on the back surface of the sheet. The sheet is then conveyed again to the pre-image-forming registration correction unit, the print belt unit **2200**, and the recording unit **2300** of the printing module **2000**, and printing is performed on the sheet.

[0064] The reversing module **6000** includes a second reversing unit **6400**. The reversing module **6000** can reverse, by the second reversing unit **6400**, the front and back surfaces of the sheet to be conveyed. Thus, the orientation of the front and back surfaces of the sheet to be discharged can be changed. The sheet discharging and stacking module **7000** includes a top tray **7200** and a stacking unit **7500**. The sheet discharging and stacking module **7000** aligns and stacks the sheets conveyed from the reversing module **6000** onto the top tray **7200** or the stacking unit **7500**.

Frame Body Configuration

[0065] As described above, the ink-jet recording apparatus **100** is divided into the sheet feeding module **1000**, the printing module **2000**, the drying module **3000**, the fixing module **4000**, the cooling module **5000**, the reversing module **6000**, and the sheet discharging and stacking module **7000**. The ink-jet recording apparatus **100** includes, for each module, a frame body for supporting an internal configuration of each module. The frame bodies have different configurations because

the frame bodies support different internal configurations, respectively. Here, the frame body of the fixing module **4000** is described.

[0066] FIG. **9** is a configuration view for illustrating an example of the frame body of the fixing module **4000**. The frame body of the fixing module **4000** is referred to as “fixing frame.” The fixing frame **400** is formed of a bottom plate frame **401**, a rear frame **402**, a right frame **403**, a left frame **404**, and a center shelf **405** that are joined together. The bottom plate frame **401**, the rear frame **402**, the right frame **403**, the left frame **404**, and the center shelf **405** are each formed of a plurality of frame members that are joined together.

[0067] Now, the configuration of the bottom plate frame **401** is described. FIG. **10** is a configuration view for illustrating an example of the bottom plate frame **401**. The bottom plate frame **401** is formed of seventeen frame members of six types including: a bottom plate **450**; two lateral beams **451**; two longitudinal beams **452**; four end diagonal beams **453**; four center diagonal beams **454**; and four center connecting beams **455**. The frame members are joined together by welding after being combined with each other.

[0068] Similarly, the rear frame **402**, the right frame **403**, the left frame **404**, and the center shelf **405** are each formed of a plurality of frame members, which are similarly joined together by welding after the frame members are combined with each other. In order to suppress deformation of the frame members due to welding, welding to be used for joining the frame members together is preferably superposition welding by laser welding, which has a relatively small amount of heat input to the frame members. In particular, welding by the above-mentioned joining method according to this embodiment is preferred. Accordingly, superposition welding by laser welding is mainly used for joining the frame members together.

[0069] FIG. **11** is an exploded view for illustrating the fixing frame **400**. The bottom plate frame **401**, the rear frame **402**, the right frame **403**, the left frame **404**, and the center shelf **405** are each produced by joining the plurality of frame members together as described above. The fixing frame **400** is produced by combining the bottom plate frame **401**, the rear frame **402**, the right frame **403**, the left frame **404**, and the center shelf **405** and joining the frames together by welding. In a case of welding the bottom plate frame **401**, the rear frame **402**, the right frame **403**, the left frame **404**, and the center shelf **405** together, superposition welding by laser welding is also preferred for the same reasons as those for welding the frame members together. In particular, welding by the above-mentioned joining method according to this embodiment is preferred. Before the joining step by welding, temporary support is preferable for maintaining a proper gap between the frames.

Temporary Support for Welding Frame Body

[0070] FIG. **12** is a configuration view for illustrating a state in which the frames of the fixing frame **400** are temporarily supported. FIG. **13** is an enlarged view for illustrating a region B of FIG. **12**. In the related-art, each frame is temporarily supported using a jig (temporary support jig) for temporarily supporting the bottom plate frame **401**, the rear frame **402**, the right frame **403**, the left frame **404**, and the center shelf **405**. However, in this embodiment, the frames are fastened to each other with the rivets **103**, and hence the fixing frame **400** can stand on its own even in a state before welding.

[0071] As illustrated in FIG. **13**, the bottom plate frame **401** has rivets **410** and joining points **411** used for joining the bottom plate frame **401**. The right frame **403** similarly has rivets **412** and joining points (not shown) used for joining the right frame **403**. To configure the fixing frame **400**, the bottom plate frame **401** and the right frame **403** are temporarily supported by the plurality of rivets **103**, and the superposed surfaces of the frames are restrained from separating from each other. Similarly, the rear frame **402**, the left frame **404**, and the center shelf **405** are also temporarily supported by the plurality of rivets **103**. Accordingly, even before the frames are joined together by welding, the surfaces of the frames that are superposed on each other near the joining points to be formed by welding do not separate from each other, and the fixing frame **400** can stand on its own.

[0072] FIG. 14 is an explanatory view for illustrating a state in which the fixing frame **400** before the joining step is mounted on a welding apparatus. The welding apparatus is a laser welding apparatus configured to include a welding robot **810** and a rotary table **811**. The fixing frame **400** to be welded is fixed on the rotary table **811**. The welding robot **810** performs welding on the frames of the fixing frame **400** fixed on the rotary table **811**.

[0073] The bottom plate frame **401** of the fixing frame **400** is directly fixed on the rotary table **811**. The fixing frame **400** can stand on its own, and hence even in a temporarily supported state, the fixing frame **400** is fixed on the rotary table **811** through fixing the bottom plate frame **401**.

[0074] In the frame body structure of the ink-jet recording apparatus **100** according to this embodiment, the bottom plate frame **401** has a substantially flat shape, and hence it is easy to fix the bottom plate frame **401** on the rotary table **811**. Further, the shape for being fixed on the rotary table **811** can be standardized for the frame body structure of each module of the ink-jet recording apparatus **100**. This eliminates the need to fix the jig for temporary support of each frame body on the rotary table **811** during the joining step of welding the frame body configuration of each module, and hence can significantly reduce the man-hours used to replace the frame body to be welded.

[0075] FIG. 15 is an enlarged view for illustrating the region B of FIG. 12 after joining (after welding). The bottom plate frame **401** and the right frame **403** are temporarily supported by the rivets **103**, each of the joining points **104** is set at a position apart by a distance of **30** mm or more and **250** mm or less from the center of the rivet **103**, and welding is performed on the joining points **104**. When each of the joining points **104** is set at a position apart by a distance of **30** mm or more and **250** mm or less from the center of the rivet **103**, a part of the steel sheet can be deformed elastically during welding, thereby allowing release of zinc vapor generated between the superposed surfaces. Further, the gap between two members (between two frames) to be welded can be set to **0.4** mm or less.

[0076] As described above, in this embodiment, a plurality of frames are temporarily supported by a joining member to produce the frame body of the image forming apparatus, thereby allowing the frame body to stand on its own. Then, the frames are joined together by performing laser welding on a position apart by a predetermined distance or more from the position of joining performed by the joining member. When the position apart by a predetermined distance or more from the joining position is set as the joining point to be formed by welding, the vapor of the plating metal is released from the gap **109** formed due to a minute change in at least one of the frames caused by elastic deformation when the vapor is generated from the plating metal. Accordingly, the occurrence of, for example, blowholes caused by the vapor of the plating metal can be suppressed, and the deterioration in welding quality can be prevented.

[0077] As described above, in the joining method according to this embodiment, unlike the related art, it is not necessary that the frame member have a step shape to release the vapor of the plating metal. Further, it is not necessary to use a jig capable of securing a ventilation channel as temporary support before the joining step by welding, or to provide a preliminary step such as removing zinc from the surface of the steel sheet by performing laser irradiation before the main welding. Accordingly, welding can be performed with significantly lower cost and less man-hours, and with less deterioration in welding quality than those of the related art.

[0078] For reference, a related-art joining method is described. FIG. 16 is an explanatory view for illustrating a related-art temporary support jig for welding a frame body. A related-art temporary support jig **800** for welding a frame body includes a bottom plate support portion **801**, a rear support portion **802**, a right support portion **803**, and a left support portion **804**. Those support portions are separable from each other, and each include a plurality of frame pressing members **805**. The bottom plate support portion **801** temporarily supports the bottom plate frame **401** with the plurality of frame pressing members **805**. The rear support portion **802** temporarily supports the rear frame **402** with the plurality of frame pressing members **805**. The right support portion **803**

temporarily supports the right frame **403** with the plurality of frame pressing members **805**. The left support portion **804** temporarily supports the left frame **404** with the plurality of frame pressing members **805**.

[0079] FIGS. **17A** to **17C** are each an explanatory view for illustrating a related-art frame joining method. The first galvanized steel sheet **101** and the second galvanized steel sheet **102**, which form a frame, are pressed by the frame pressing members **805**, and thus are prevented from separating from each other (FIG. **17A**). The frame pressing member **805** prevents the separation between the steel sheets by urging a pressing portion **807** with a pressing spring **808** in a direction opposite to a support portion **806**. A clearance between the pressing portion **807** and a frame receiving surface **809** is set to a value smaller than the sum of the sheet thickness of the first galvanized steel sheet **101** and the sheet thickness of the second galvanized steel sheet **102**. For this reason, the pressing portion **807** presses the frame receiving surface **809** so as to prevent the two steel sheets from separating from each other. The vicinity of the frame pressing member **805** is irradiated with the laser beam **107**. The gap formed between the first galvanized steel sheet **101** and the second galvanized steel sheet **102** is configured to be **0.4** mm or less.

[0080] The laser beam **107** forms the molten pool **108** in the first galvanized steel sheet **101** and the second galvanized steel sheet **102** (FIG. **17B**). The zinc vapor generated between the steel sheets pushes up the first galvanized steel sheet **101**, thereby causing the pressing spring **808** to contract slightly. Owing to the contraction of the pressing spring **808**, the gap **109** is formed between the first galvanized steel sheet **101** and the second galvanized steel sheet **102**. The zinc vapor is released from the gap **109** as the exhaust air **110**. When welding by the laser beam **107** is completed, on the second galvanized steel sheet **102**, the protrusion **111** is formed as a mark of laser welding (FIG. **17C**).

[0081] FIG. **18** is an explanatory view for illustrating a state in which the temporary support jig **800** supports each frame therein. The temporary support jig **800** for welding a frame body has the large-scale configuration because the temporary support jig **800** for welding a frame body has such a configuration as to cover a frame body in terms of the structure. FIG. **19** is an explanatory view for illustrating a state in which the temporary support jig **800** is mounted on the welding apparatus. When welding is performed on the fixing frame **400**, the temporary support jig **800** is mounted on the rotary table **811** in order to perform welding from a plurality of directions. The temporary support jig **800** for welding a frame body is a large-sized jig, and hence the welding robot **810** and the rotary table **811** are also preferred to be large.

[0082] The ink-jet recording apparatus **100** may use seven types of frame bodies with different configurations. Accordingly, the temporary support jig **800** for welding a frame body is preferable for each of the frame bodies. Further, it is preferable to sequentially mount the large-scale temporary support jigs **800** for welding a frame body on the welding apparatus so as to perform welding, and hence the man-hours used for the joining step by welding are increased. When the joining method according to this embodiment is applied, the above-mentioned related-art temporary support jig **800** is not needed, and thus the man-hours used for the welding step can be significantly reduced as compared to the related art. Thus, according to the present disclosure, even in the frame body configuration in which a large number of metal sheets are mounted from a plurality of directions, it is possible to perform welding on the metal sheets while securing a ventilation channel for releasing metal vapor.

[0083] While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0084] This application claims the benefit of priority from Japanese Patent Application No. 2024-025599, filed Feb. 22, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. A joining method of joining a first metal sheet and a second metal sheet, the joining method comprising: a fastening step of fastening the first metal sheet and the second metal sheet, which have been superposed on each other, by a fastening member at a fastening position; and a welding step of welding the first metal sheet and the second metal sheet, which have been superposed on each other, at a welding position, wherein, in a direction intersecting a sheet thickness direction of the first metal sheet, a distance between the fastening position and the welding position is set such that vapor generated between the first metal sheet and the second metal sheet during the welding step elastically deforms at least one of the first metal sheet and the second metal sheet to form a gap between the first metal sheet and the second metal sheet.
2. The joining method according to claim 1, wherein at least one of the first metal sheet and the second metal sheet includes: a base metal, and a coating material that has a boiling point lower than a melting point of the base metal, and coats the base metal.
3. The joining method according to claim 1, wherein the gap does not exceed 0.4 mm (millimeter) during a period in which the welding step is performed.
4. The joining method according to claim 1, wherein the first sheet metal is deformed by the vapor to a larger extent than the second metal is, and wherein the distance is at least 9.3 times a sheet thickness of the first metal sheet.
5. The joining method according to claim 4, wherein the first metal sheet and the second metal sheet each have a sheet thickness of from 0.8 mm (millimeter) to 3.2 mm (millimeter).
6. The joining method according to claim 1, wherein the distance is 30 mm (millimeter) or more.
7. The joining method according to claim 1, wherein the distance is 250 mm (millimeter) or less.
8. The joining method according to claim 1, wherein the fastening member is a rivet.
9. A joining method of joining a first metal sheet and a second metal sheet, the joining method comprising: a fastening step of fastening the first metal sheet and the second metal sheet, which have been superposed on each other, by a fastening member at a fastening position; and a welding step of welding the first metal sheet and the second metal sheet, which have been superposed on each other, at a welding position, wherein a distance between the fastening position and the welding position in a direction intersecting a sheet thickness direction of the first metal sheet is 30 mm (millimeter) or more.
10. The joining method according to claim 9, wherein at least one of the first metal sheet and the second metal sheet includes: a base metal; and a coating material that has a boiling point lower than a melting point of the base metal, and coats the base metal.
11. The joining method according to claim 9, wherein vapor generated between the first metal sheet and the second metal sheet during the welding step elastically deforms at least one of the first metal sheet and the second metal sheet to form a gap between the first metal sheet and the second metal sheet, and wherein the gap does not exceed 0.4 mm (millimeter) during a period in which the welding step is performed.
12. The joining method according to claim 11, wherein the first sheet metal is deformed by the vapor to a larger extent than the second metal is, and wherein the distance is at least 9.3 times a sheet thickness of the first metal sheet.
13. The joining method according to claim 11, wherein the first metal sheet and the second metal sheet each have a sheet thickness of from 0.8 mm (millimeter) to 3.2 mm (millimeter).
14. The joining method according to claim 9, wherein the distance is 250 mm (millimeter) or less.
15. The joining method according to claim 9, wherein the fastening member is a rivet.
16. A manufacturing method of an image forming apparatus, comprising: preparing a frame body using the joining method according to claim 1; and providing an image forming unit to the frame body.

17. A manufacturing method of an image forming apparatus, comprising: preparing a frame body using the joining method according to claim 9; and providing an image forming unit to the frame body.

18. An image forming apparatus, comprising: an image forming unit; and a frame body, which includes a first metal sheet and a second metal sheet joined together, and supports the image forming unit, wherein the first metal sheet and the second metal sheet are fastened together by a fastening member at a fastening position, wherein the first metal sheet and the second metal sheet are welded together at a welding position, and wherein a distance between the fastening position and the welding position in a direction intersecting a sheet thickness direction of the first metal sheet is 30 mm (millimeter) or more.

19. The image forming apparatus according to claim 18, wherein at least one of the first metal sheet and the second metal sheet includes: a base metal; and a coating material that has a boiling point lower than a melting point of the base metal, and coats the base metal.

20. The image forming apparatus according to claim 18, wherein the distance is 250 mm (millimeter) or less.
